

# simple-plot

Typst Package

*Mathematical Function Plotting*

---

A lightweight library for creating elegant mathematical plots

Version 0.9.1

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# 1. Introduction

`simple-plot` is a Typst package for creating clean, elegant mathematical plots. Built on CeTZ, it provides an intuitive interface for plotting functions, data points, and creating publication-ready graphs.

## 1.1. Features

- Plot mathematical functions with automatic sampling
- Parametric curves (ellipses, hyperbolas, closed shapes)
- Scatter plots and line plots with customizable markers
- Fill areas under curves or between two curves (solid or hatched)
- Clean integer-based tick system by default
- Major and minor grid with elegant styling
- Gap-based grid line breaks around tick labels (`grid-label-break`)
- Automatic axis extension beyond grid
- Flexible axis positioning (origin, bottom/left, custom)
- Multiple label display options (unit-label-only, label-step)
- Function labels with flexible positioning
- Text annotations and reference lines (`vline`, `hline`)
- Riemann sum rectangles with left, right, midpoint, lower, and upper methods
- Volume of revolution diagrams with end caps, disks, and tilted axes
- Clipping for clean rendering at boundaries
- Global defaults and style overrides

## 1.2. Installation

Import the package in your Typst document:

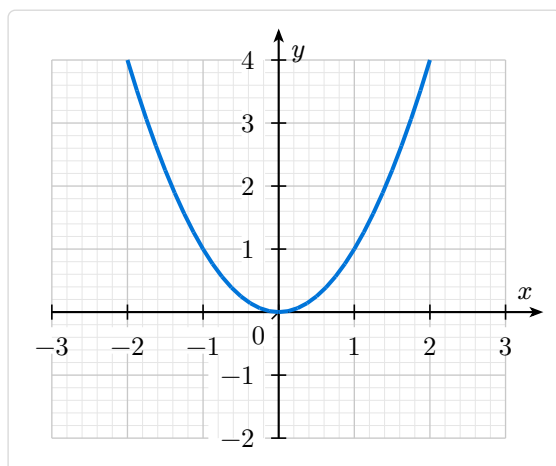
```
#import "@preview/simple-plot:1.0.0": plot
```

## 1.3. Quick Start

Code

```
#plot(  
  width: 6, height: 5,  
  xmin: -3, xmax: 3, ymin: -2, ymax: 4,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  show-grid: true,  
  (fn:  $x \Rightarrow x * x$ , stroke: blue + 1.5pt),  
)
```

Preview



## 2. Basic Usage

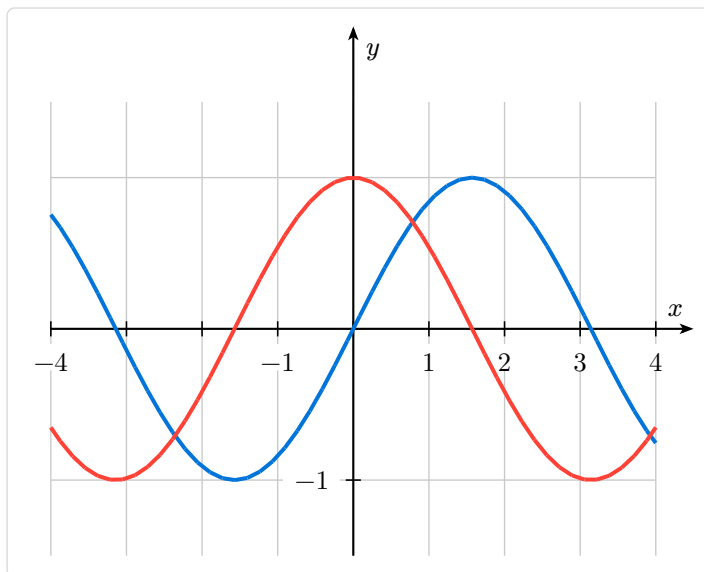
### 2.1. Plotting Functions

Plot mathematical functions by passing a dictionary with `fn`:

Code

```
#plot(  
  width: 8, height: 6,  
  xmin: -4, xmax: 4, ymin: -1.5, ymax: 1.5,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  show-grid: "major",  
  (fn: x => calc.sin(x), stroke: blue + 1.5pt),  
  (fn: x => calc.cos(x), stroke: red + 1.5pt),  
)
```

Preview



### 2.2. Mathematical Functions Reference

Functions are defined using Typst's `calc` module. Here are the most common mathematical functions:

| Function               | Typst syntax                      |
|------------------------|-----------------------------------|
| Power $x^n$            | <code>calc.pow(x, n)</code>       |
| Square root $\sqrt{x}$ | <code>calc.sqrt(x)</code>         |
| Absolute value $ x $   | <code>calc.abs(x)</code>          |
| Sine $\sin(x)$         | <code>calc.sin(x)</code>          |
| Cosine $\cos(x)$       | <code>calc.cos(x)</code>          |
| Tangent $\tan(x)$      | <code>calc.tan(x)</code>          |
| Exponential $e^x$      | <code>calc.exp(x)</code>          |
| Natural log $\ln(x)$   | <code>calc.ln(x)</code>           |
| Log base $b$           | <code>calc.log(x, base: b)</code> |
| Maximum                | <code>calc.max(a, b)</code>       |
| Minimum                | <code>calc.min(a, b)</code>       |

**Important:** When using constants in calculations, use decimal notation (e.g., 2.0 instead of 2) to avoid type errors. For example:

- ✓  $x \Rightarrow x * x / 2.0$
- ✗  $x \Rightarrow x * x / 2$  (*may cause errors*)

This is because Typst's type system requires consistent float arithmetic.

To draw a function with a hole or discontinuity, return `none` from the function at that point — the line will break without connecting across the gap:

```
// Draws 1/x with a break at x=0
(fn: x => if calc.abs(x) < 0.01 { none } else { 1.0 / x }, stroke: blue + 1.5pt)
```

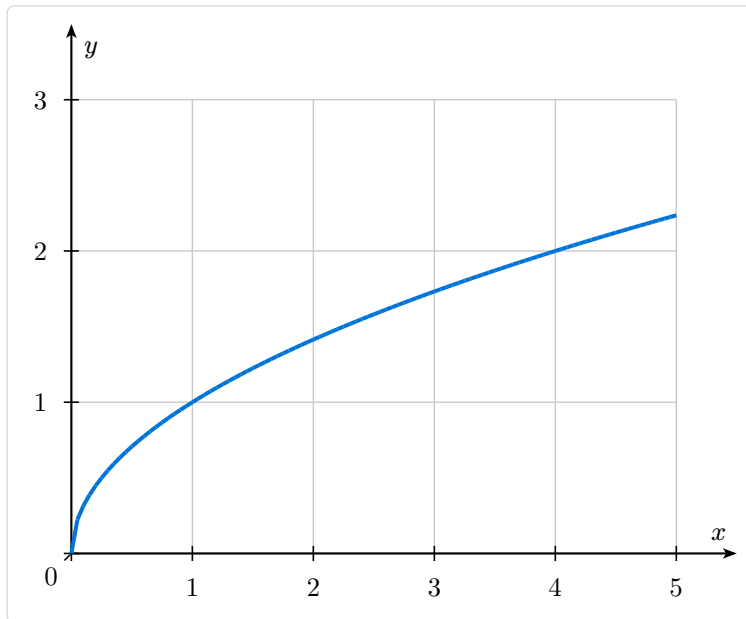
## 2.3. Function Domain

Specify a custom domain for functions:

Code

```
#plot(  
  width: 8, height: 6,  
  xmin: 0, xmax: 5, ymin: 0, ymax: 3,  
  xlabel: "$x$", ylabel: "$y$",  
  axis-x-pos: "bottom", axis-y-pos: "left",  
  show-grid: "major",  
  (fn: x => calc.sqrt(x), domain: (0, 5), stroke: blue + 1.5pt),  
)
```

Preview



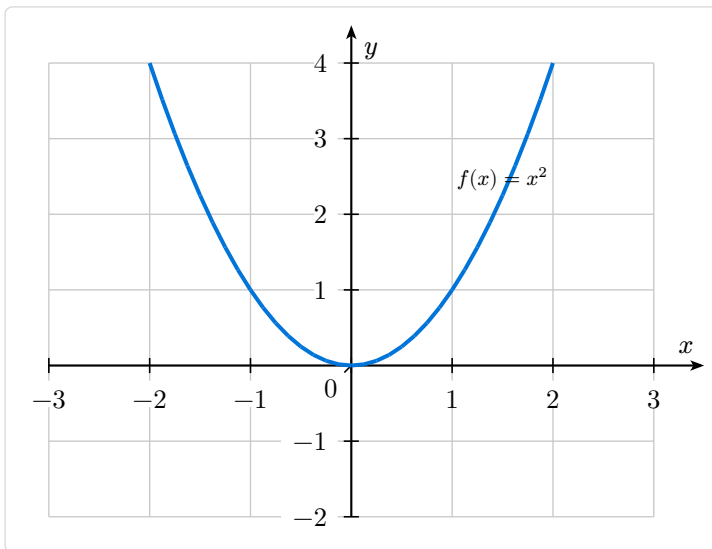
## 2.4. Function Labels

Add a `label` field to any function spec. By default, labels are placed **automatically**: the engine scores candidate positions along the curve and around each point, avoiding other curves, tick labels, axes, and previously placed labels. Set `label-anchor` or `label-side` to override placement.

Code

```
#plot(  
  width: 8, height: 6,  
  xmin: -3, xmax: 3, ymin: -2, ymax: 4,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  show-grid: "major",  
  (  
    fn: x => x * x,  
    stroke: blue + 1.5pt,  
    label:  $f(x) = x^2$ ,  
    label-side: "above",  
    label-pos: 0.75,  
  ),  
)
```

Preview



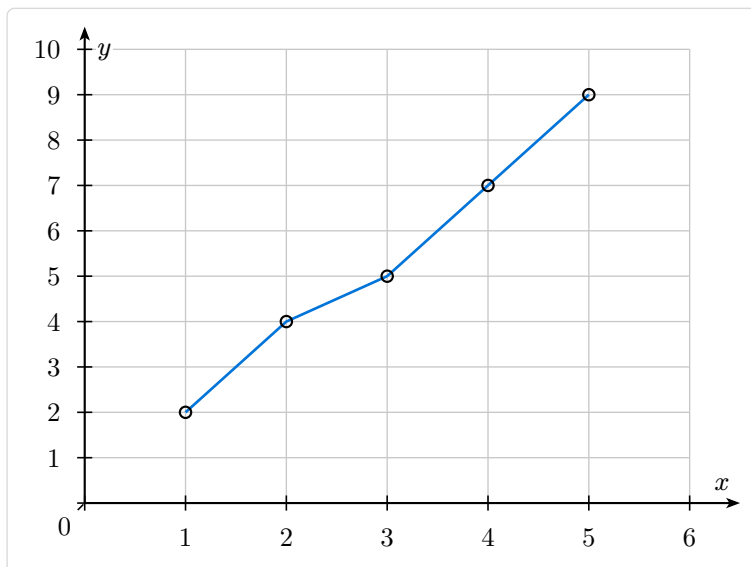
## 2.5. Data Points and Scatter Plots

Plot discrete data points:

## Code

```
#plot(  
  width: 8, height: 6,  
  xmin: 0, xmax: 6, ymin: 0, ymax: 10,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  axis-x-pos: "bottom", axis-y-pos: "left",  
  show-grid: "major",  
  (  
    data: ((1, 2), (2, 4), (3, 5), (4, 7), (5, 9)),  
    mark: "o",  
    mark-size: 0.15,  
    stroke: blue + 1pt,  
  ),  
)
```

## Preview



## 3. Grid Options

### 3.1. Grid Modes

Control grid display with `show-grid`:

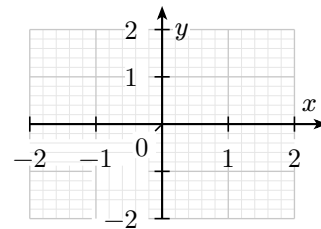
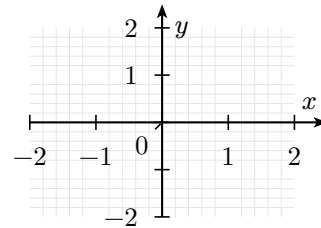
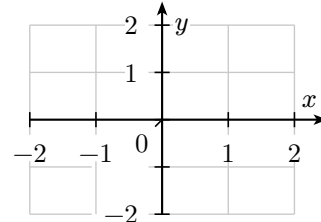
Code

```
// Major grid only
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  show-grid: "major",
)

// Minor grid only
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  show-grid: "minor",
)

// Both grids
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  show-grid: "both",
)
```

Preview



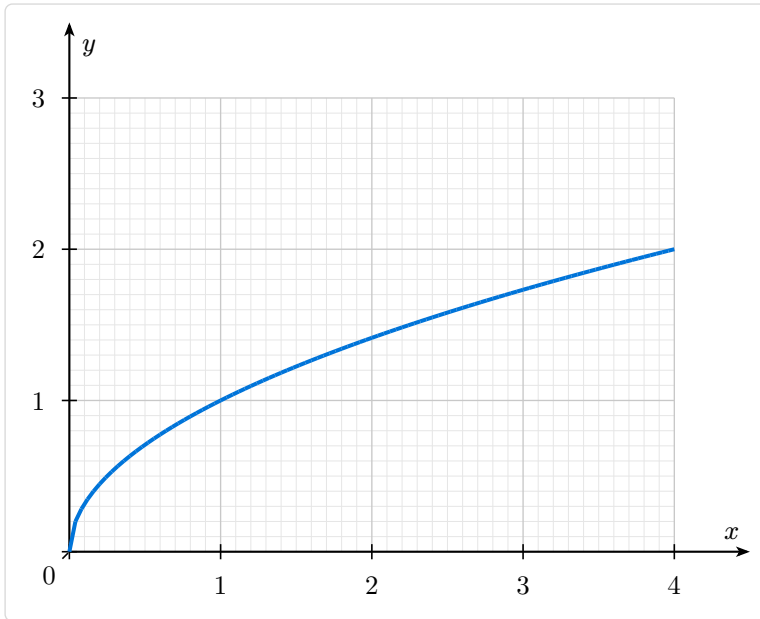
### 3.2. Minor Grid Subdivisions

Control the number of subdivisions with `minor-grid-step`:

## Code

```
#plot(  
  width: 8, height: 6,  
  xmin: 0, xmax: 4, ymin: 0, ymax: 3,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  axis-x-pos: "bottom", axis-y-pos: "left",  
  show-grid: "both",  
  minor-grid-step: 10, // 10 subdivisions per unit  
  (fn: x => calc.sqrt(x), domain: (0, 4), stroke: blue + 1.5pt),  
)
```

## Preview



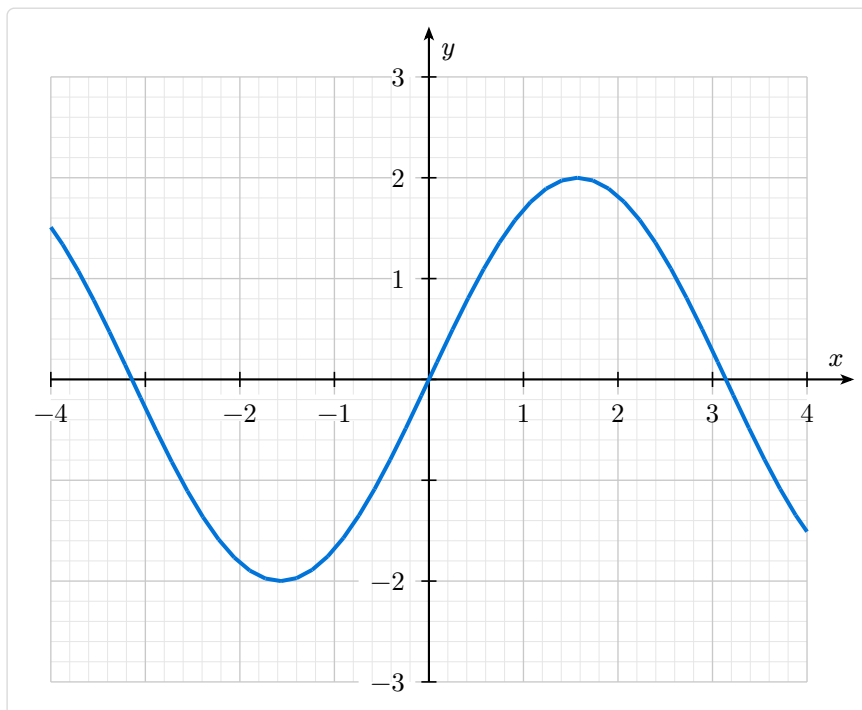
### 3.3. Grid Label Break

The `grid-label-break` option (enabled by default) draws grid lines with gaps around tick labels, creating an elegant break effect. Unlike a white-box approach, this works on any background color:

#### Code

```
#plot(
  width: 10, height: 8,
  xmin: -4, xmax: 4, ymin: -3, ymax: 3,
  xlabel:  $x$ , ylabel:  $y$ ,
  show-grid: "both",
  minor-grid-step: 5,
  grid-label-break: true, // Default
  (fn: x => calc.sin(x) * 2, stroke: blue + 1.5pt),
)
```

#### Preview



## 4. Axis Configuration

### 4.1. Axis Position

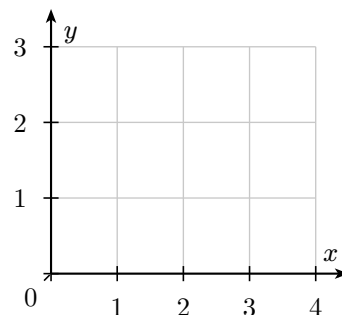
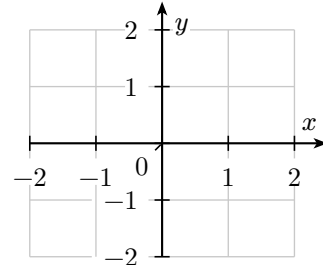
Position axes at origin (default), bottom/left, or custom values:

Code

```
// Through origin (default)
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  show-grid: "major",
)

// Bottom and left
#plot(
  width: 5, height: 4,
  xmin: 0, xmax: 4, ymin: 0, ymax: 3,
  axis-x-pos: "bottom",
  axis-y-pos: "left",
  show-grid: "major",
)
```

Preview



### 4.2. Axis Extension

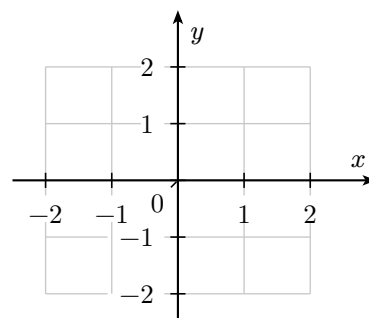
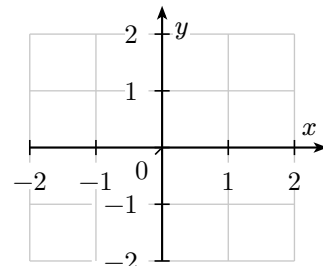
By default, axes extend 0.5 units beyond the grid on the arrow side. Customize with `axis-x-extend` and `axis-y-extend`:

Code

```
// Default extension (0, 0.5)
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  show-grid: "major",
)

// Custom extension
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  axis-x-extend: (0.5, 1),
  axis-y-extend: (0.5, 1),
  show-grid: "major",
)
```

Preview



## 5. Tick Configuration

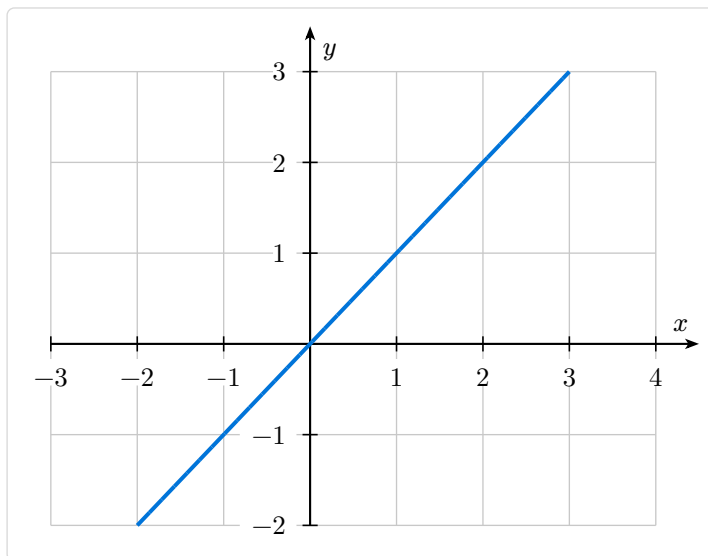
### 5.1. Default Integer Ticks

By default, ticks are placed at every integer (step = 1):

Code

```
#plot(  
  width: 8, height: 6,  
  xmin: -3, xmax: 4, ymin: -2, ymax: 3,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  show-grid: "major",  
  (fn: x => x, stroke: blue + 1.5pt),  
)
```

Preview



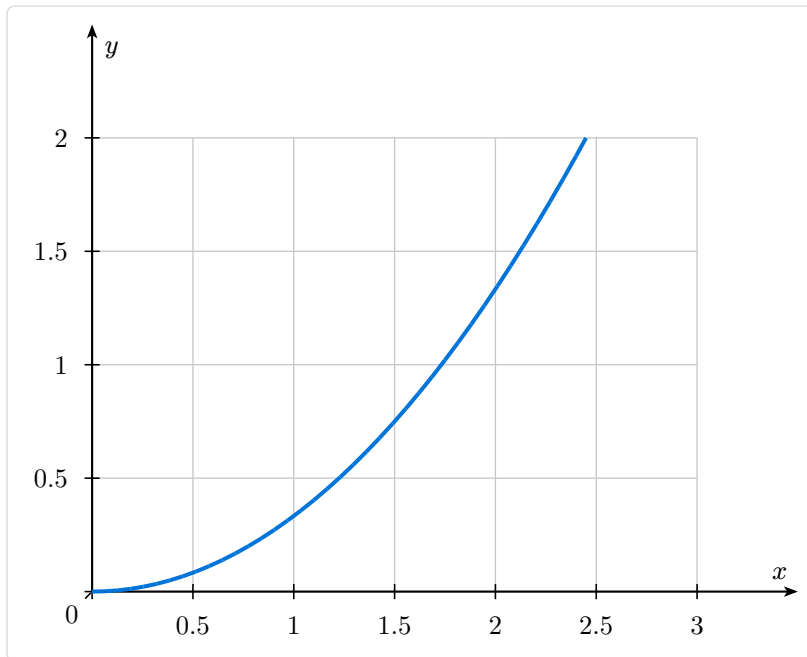
### 5.2. Custom Tick Step

Change tick spacing with `xtick-step` and `ytick-step`:

## Code

```
#plot(  
  width: 8, height: 6,  
  xmin: 0, xmax: 3, ymin: 0, ymax: 2,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  axis-x-pos: "bottom", axis-y-pos: "left",  
  xtick-step: 0.5,  
  ytick-step: 0.5,  
  show-grid: "major",  
  (fn:  $x \Rightarrow x * x / 3$ , stroke: blue + 1.5pt),  
)
```

## Preview



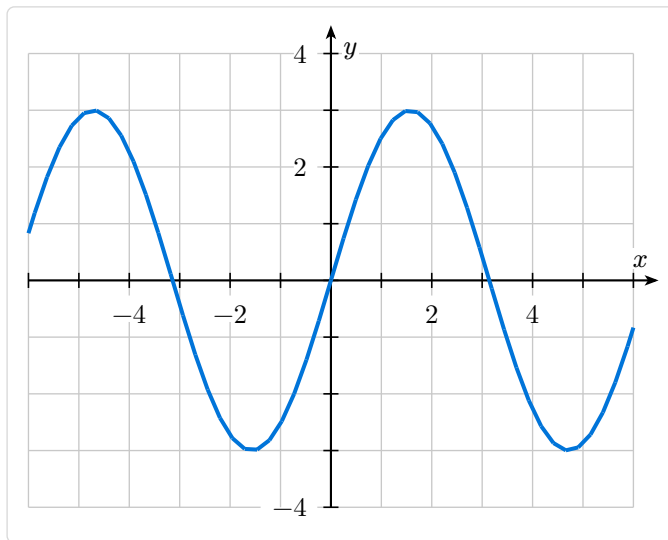
### 5.3. Tick Label Step

Show labels only at every N-th tick with `xtick-label-step` and `ytick-label-step`:

Code

```
#plot(
  width: 8, height: 6,
  xmin: -6, xmax: 6, ymin: -4, ymax: 4,
  xlabel:  $x$ , ylabel:  $y$ ,
  xtick-label-step: 2, // Labels at -6, -4, -2, 2, 4, 6
  ytick-label-step: 2, // Labels at -4, -2, 2, 4
  show-grid: "major",
  (fn: x => calc.sin(x) * 3, stroke: blue + 1.5pt),
)
```

Preview



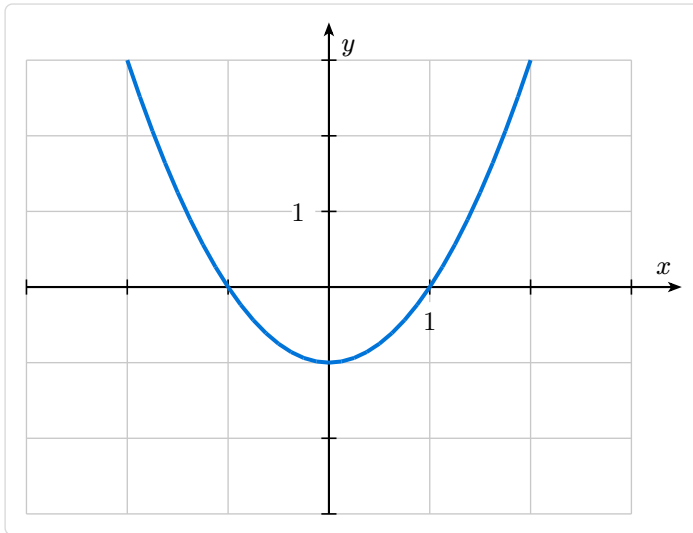
### 5.4. Unit Label Only

Show only “1” on each axis for a minimal style:

## Code

```
#plot(  
  width: 8, height: 6,  
  xmin: -3, xmax: 3, ymin: -3, ymax: 3,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  unit-label-only: true,  
  show-origin: false,  
  show-grid: "major",  
  (fn: x => x * x - 1, stroke: blue + 1.5pt),  
)
```

## Preview



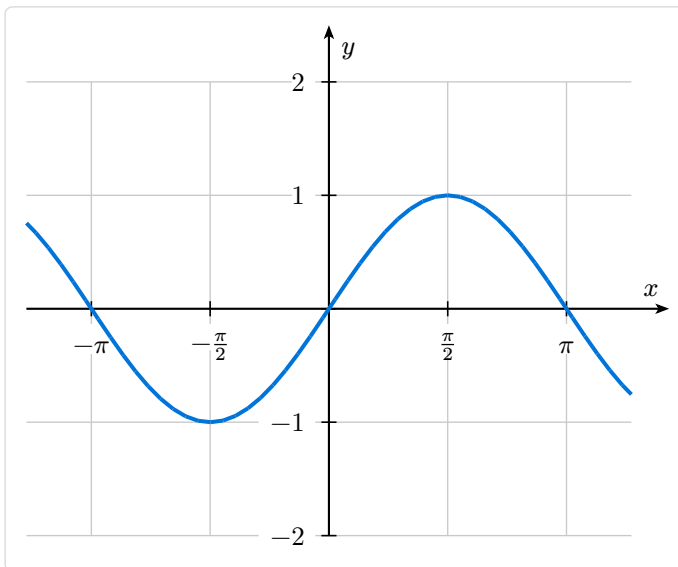
## 5.5. Custom Tick Positions

Specify exact tick positions with `xtick` and `ytick`:

Code

```
#plot(
  width: 8, height: 6,
  xmin: -4, xmax: 4, ymin: -2, ymax: 2,
  xlabel:  $x$ , ylabel:  $y$ ,
  xtick: (-calc.pi, -calc.pi/2, 0, calc.pi/2, calc.pi),
  xtick-labels: ( $-\pi$ ,  $-\pi/2$ ,  $0$ ,  $\pi/2$ ,  $\pi$ ),
  show-grid: "major",
  (fn: x => calc.sin(x), stroke: blue + 1.5pt),
)
```

Preview



## 5.6. Hide Origin Label

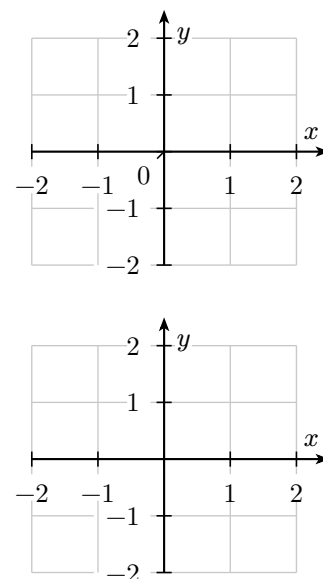
Control the “0” label at the origin:

Code

```
// With origin (default)
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  show-origin: true,
  show-grid: "major",
)

// Without origin
#plot(
  width: 5, height: 4,
  xmin: -2, xmax: 2, ymin: -2, ymax: 2,
  show-origin: false,
  show-grid: "major",
)
```

Preview



## 6. Markers

### 6.1. Available Marker Types

The following markers are available:

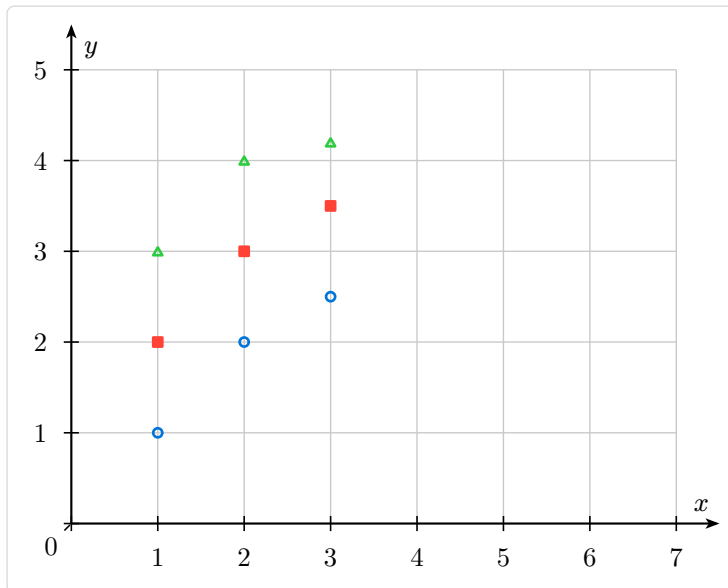
| Marker      | Description           |
|-------------|-----------------------|
| "o"         | Circle (outline)      |
| "*"         | Circle (filled)       |
| "square"    | Square (outline)      |
| "square*"   | Square (filled)       |
| "triangle"  | Triangle up (outline) |
| "triangle*" | Triangle up (filled)  |
| "diamond"   | Diamond (outline)     |
| "diamond*"  | Diamond (filled)      |
| "star"      | Star (outline)        |
| "star*"     | Star (filled)         |
| "+"         | Plus sign             |
| "x"         | Cross                 |
| " "         | Vertical bar          |
| "_"         | Horizontal bar        |

## 6.2. Using Markers

Code

```
#plot(  
  width: 8, height: 6,  
  xmin: 0, xmax: 7, ymin: 0, ymax: 5,  
  axis-x-pos: "bottom", axis-y-pos: "left",  
  show-grid: "major",  
  data(((1, 1), (2, 2), (3, 2.5))), mark: "o", mark-stroke: blue),  
  data(((1, 2), (2, 3), (3, 3.5))), mark: "square*", mark-fill: red, mark-stroke: red),  
  data(((1, 3), (2, 4), (3, 4.2))), mark: "triangle", mark-stroke: green),  
)
```

Preview



## 7. Convenience Functions

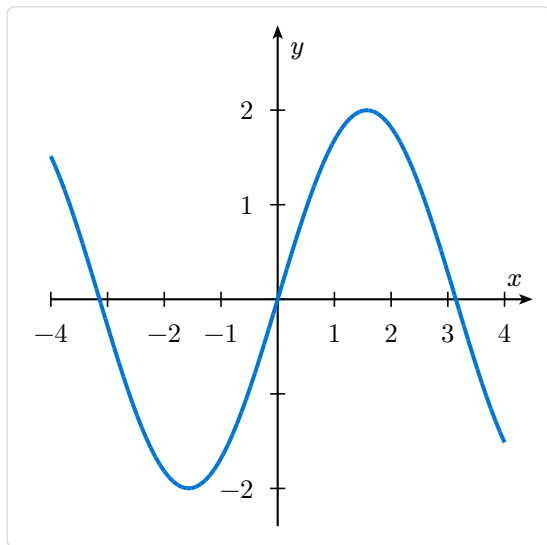
### 7.1. plot-fn - Quick Function Plot

Plot a single function with automatic y-scaling:

Code

```
#plot-fn(  
  x => calc.sin(x) * 2.0,  
  domain: (-4, 4),  
  stroke: blue + 1.5pt,  
)
```

Preview



Parameters:

| Parameter   | Type       | Default      | Description                                   |
|-------------|------------|--------------|-----------------------------------------------|
| fn          | function   | —            | Function to plot                              |
| domain      | array      | (-5, 5)      | X domain                                      |
| ymin / ymax | float/auto | auto         | Y bounds; <b>auto</b> = computed from samples |
| stroke      | stroke     | blue + 1.2pt | Line style                                    |
| samples     | int        | 100          | Sample count (auto y-range and curve)         |
| ..args      | any        | —            | Forwarded to <code>plot()</code>              |

### 7.2. plot-rational - Rational Function Wrapper

`plot-rational` is a convenience wrapper for common rational-function exercise graphs. It draws a single function on a centered coordinate grid and forwards extra arguments to `plot`, so you can add horizontal asymptotes, holes, or labels. Pass vertical asymptotes with `vertical-asymptotes`; the function domain is split around those values to avoid sampling at the pole.

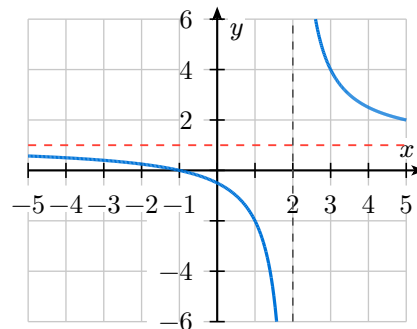
```
#plot-rational(  
  x => (x + 1) / (x - 2),  
  xmin: -5, xmax: 5,  
  ymin: -6, ymax: 6,  
  vertical-asymptotes: (2,),
```

```
hline(1, stroke: stroke(paint: red, thickness: 0.7pt, dash: "dashed")),
)
```

## Code

```
#plot-rational(
  x => (x + 1) / (x - 2),
  xmin: -5, xmax: 5,
  ymin: -6, ymax: 6,
  vertical-asymptotes: (2,),
  hline(1, stroke: stroke(paint: red, thickness: 0.7pt,
dash: "dashed")),
)
```

## Preview



## Parameters:

| Parameter           | Type       | Default      | Description                                              |
|---------------------|------------|--------------|----------------------------------------------------------|
| fn                  | function   | —            | Function to plot                                         |
| xmin / xmax         | float      | -6 / 6       | X-axis bounds                                            |
| ymin / ymax         | float      | -8 / 8       | Y-axis bounds                                            |
| stroke              | stroke     | blue + 1.2pt | Function stroke                                          |
| domain              | array/auto | auto         | Function domain; auto = (xmin, xmax)                     |
| samples             | int        | 200          | Function sample count                                    |
| vertical-asymptotes | array      | ()           | X-values to skip and draw as vertical dashed lines       |
| asymptote-stroke    | stroke     | dashed gray  | Stroke for vertical asymptote lines                      |
| asymptote-gap       | float      | 0.05         | Half-gap removed around each vertical asymptote          |
| width / height      | float/auto | auto         | Plot dimensions                                          |
| ..args              | any        | —            | Forwarded to plot(); positional args become extra series |

## 7.3. limit-schema / schema-lim - Local Limit Sketches

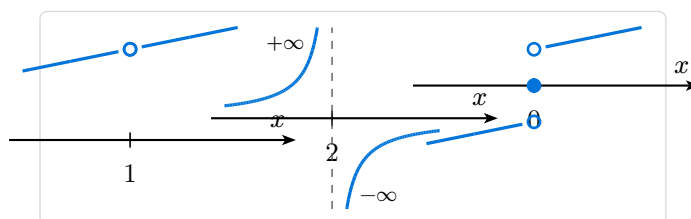
limit-schema draws a small schematic graph around a point **a**. It is useful for removable discontinuities, jumps, and vertical asymptotes. schema-lim is kept as a French alias.

```
#limit-schema(a: 1, left: 4, right: 4)
#limit-schema(a: 2, left: "+oo", right: "-oo")
#limit-schema(a: 0, left: -1, right: 1, val: 0)
```

## Code

```
#grid(columns: (1fr, 1fr, 1fr), gutter: 0.5em,
  limit-schema(a: 1, left: 4, right: 4),
  limit-schema(a: 2, left: "+∞", right: "-∞"),
  limit-schema(a: 0, left: -1, right: 1, val: 0),
)
```

## Preview



## Parameters:

| Parameter      | Type                  | Default   | Description                                                                         |
|----------------|-----------------------|-----------|-------------------------------------------------------------------------------------|
| a              | float                 | 0         | X-position of the local behavior                                                    |
| left           | float/string/<br>none | none      | Left-hand behavior: finite value, "+∞", "-∞", or none                               |
| right          | float/string/<br>none | none      | Right-hand behavior: finite value, "+∞", "-∞", or none                              |
| val            | float/none            | none      | Defined value $f(a)$ as a filled dot                                                |
| a-label        | content/auto          | auto      | Tick label at <b>a</b> ; <b>auto</b> prints the value                               |
| show-limit     | bool/float            | false     | Draw a dashed horizontal limit line; <b>true</b> auto-detects a common finite limit |
| l-label        | content/auto          | auto      | Label for the limit line                                                            |
| width / height | float                 | 3.8 / 2.8 | Schema dimensions in cm                                                             |

## 7.4. scatter / data - Point Sets

`scatter` and `data` are identical functions. Use them to create isolated point specifications. Set `connect: true` to join points with a line.

```
#plot(
  xmin: 0, xmax: 4, ymin: 0, ymax: 6,
  scatter(((1, 2), (2, 4), (3, 5))), mark: "o", mark-fill: blue),
  data(((1, 1.5), (2, 3.5), (3, 4.5))), mark: "*", mark-fill: red),
)
```

## Parameters (both functions):

| Parameter   | Type    | Default      | Description                                 |
|-------------|---------|--------------|---------------------------------------------|
| points      | array   | —            | Array of (x, y) tuples                      |
| mark        | string  | "*"          | Marker type                                 |
| mark-size   | float   | 0.12         | Marker size in cm                           |
| mark-fill   | color   | blue         | Marker fill color                           |
| mark-stroke | stroke  | blue + 0.8pt | Marker stroke                               |
| connect     | bool    | false        | Join points with a line                     |
| stroke      | stroke  | none         | Line stroke when <code>connect: true</code> |
| label       | content | none         | Series label                                |
| label-pos   | float   | 0.8          | Position along series in [0, 1]             |

|              |        |              |              |
|--------------|--------|--------------|--------------|
| label-anchor | string | “south-west” | Label anchor |
|--------------|--------|--------------|--------------|

## 7.5. line-plot - Connected Line Plot

Create a line plot with markers (points joined by default):

```
#plot(
  xmin: 0, xmax: 4, ymin: 0, ymax: 6,
  line-plot(((1, 2), (2, 4), (3, 5))), stroke: blue + 1pt, mark: "o"),
)
```

Parameters:

| Parameter    | Type    | Default      | Description                     |
|--------------|---------|--------------|---------------------------------|
| points       | array   | —            | Array of (x, y) tuples          |
| stroke       | stroke  | blue + 1.2pt | Line stroke                     |
| mark         | string  | "o"          | Marker type                     |
| mark-size    | float   | 0.1          | Marker size in cm               |
| mark-fill    | color   | white        | Marker fill                     |
| mark-stroke  | stroke  | blue + 0.8pt | Marker stroke                   |
| label        | content | none         | Series label                    |
| label-pos    | float   | 0.8          | Position along series in [0, 1] |
| label-anchor | string  | “south-west” | Label anchor                    |

## 7.6. func-plot - Function Plot Helper

Build a function series spec with full marker and label control:

```
#let my-func = func-plot(
  x => calc.sin(x),
  stroke: blue + 1.5pt,
  label: $sin(x)$,
  mark: "o",
  mark-interval: 15, // marker every 15th sample
)
#plot(xmin: -4, xmax: 4, ymin: -1.5, ymax: 1.5, my-func)
```

Parameters:

| Parameter   | Type       | Default      | Description                          |
|-------------|------------|--------------|--------------------------------------|
| fn          | function   | —            | Function to plot (positional or fn:) |
| domain      | array/auto | auto         | X domain; auto = full plot range     |
| stroke      | stroke     | blue + 1.2pt | Line style                           |
| samples     | int        | 100          | Sample count                         |
| mark        | string     | "none"       | Marker type                          |
| mark-size   | float      | 0.1          | Marker size in cm                    |
| mark-fill   | color      | blue         | Marker fill                          |
| mark-stroke | stroke     | blue + 0.8pt | Marker stroke                        |

|               |         |              |                                 |
|---------------|---------|--------------|---------------------------------|
| mark-interval | int     | 10           | Draw a marker every N-th sample |
| label         | content | none         | Curve label                     |
| label-pos     | float   | 0.8          | Position in [0, 1]              |
| label-anchor  | string  | “south-west” | Label anchor                    |

## 7.7. parametric - Ellipses and Hyperbolas

Draw curves that are not single-valued functions of  $x$ , such as ellipses:

```
#plot(
  xmin: -3, xmax: 3, ymin: -2, ymax: 2,
  parametric(t => 2 * calc.cos(t), t => calc.sin(t), domain: (0, 2 * calc.pi)),
)
```

For a hyperbola branch, use a finite parameter interval:

```
#plot(
  xmin: -4, xmax: 4, ymin: -3, ymax: 3,
  parametric(t => (calc.exp(t) + calc.exp(-t)) / 2, t => (calc.exp(t) - calc.exp(-t)) / 2,
  domain: (-1.5, 1.5)),
  parametric(t => -(calc.exp(t) + calc.exp(-t)) / 2, t => (calc.exp(t) - calc.exp(-t)) / 2,
  domain: (-1.5, 1.5)),
)
```

Parameters:

| Parameter | Type     | Default      | Description                     |
|-----------|----------|--------------|---------------------------------|
| fn-x      | function | —            | X coordinate as function of $t$ |
| fn-y      | function | —            | Y coordinate as function of $t$ |
| domain    | array    | (0.0, 1.0)   | Parameter range $(t_1, t_2)$    |
| stroke    | stroke   | blue + 1.2pt | Curve stroke                    |
| samples   | int      | 100          | Number of parameter samples     |

## 7.8. fill-area - Fill Under a Curve

Fill the region between a function and a baseline (default:  $y = 0$ ):

```
#plot(
  xmin: 0, xmax: calc.pi, ymin: -0.2, ymax: 1.2,
  fill-area(x => calc.sin(x), domain: (0, calc.pi), color: blue.lighten(70%)),
  (fn: x => calc.sin(x), domain: (0, calc.pi), stroke: blue + 1.5pt),
)
```

Use `baseline` for a non-zero base level, or `hatch` for hatched fills — either the flat form or a single dict:

```
fill-area(x => calc.sin(x), domain: (0, calc.pi),
  hatch: "ne", hatch-spacing: 5pt, hatch-stroke: blue + 0.5pt)
// equivalent:
fill-area(x => calc.sin(x), domain: (0, calc.pi),
  hatch: (style: "ne", spacing: 5pt, stroke: blue + 0.5pt))
```

Parameters:

| Parameter     | Type          | Default          | Description                                                                                              |
|---------------|---------------|------------------|----------------------------------------------------------------------------------------------------------|
| fn            | function      | —                | Function $f(x)$ bounding the top of the region (positional or fn:)                                       |
| domain        | array/auto    | auto             | X interval; auto = full plot range                                                                       |
| baseline      | float         | 0.0              | Y-value of the bottom of the region                                                                      |
| color         | color         | luma(220)        | Fill color; none for hatch-only                                                                          |
| hatch         | str/dict/none | none             | Hatch pattern: "ne", "nw", "h", "v", "cross", "grid" — or (style: "ne", spacing: 5pt, stroke: red + 1pt) |
| hatch-spacing | length        | 5pt              | Spacing between hatch lines                                                                              |
| hatch-stroke  | stroke        | luma(80) + 0.5pt | Hatch line stroke                                                                                        |
| samples       | int           | 80               | Sample count                                                                                             |

## 7.9. area-between - Fill Between Two Curves

Fill the region enclosed by two functions:

```
#plot(
  xmin: -1, xmax: 3, ymin: -1, ymax: 10,
  area-between(x => x * x, x => 3 * x, domain: (0, 3),
    color: green.lighten(60%)),
  (fn: x => x * x, stroke: blue + 1.5pt),
  (fn: x => 3 * x, stroke: red + 1.5pt),
)
```

Parameters:

| Parameter     | Type          | Default          | Description                                                                  |
|---------------|---------------|------------------|------------------------------------------------------------------------------|
| fn1           | function      | —                | First bounding function (positional or fn1:)                                 |
| fn2           | function      | —                | Second bounding function (positional or fn2:)                                |
| domain        | array/auto    | auto             | X interval; auto = full plot range                                           |
| color         | color         | luma(220)        | Fill color; none for hatch-only                                              |
| hatch         | str/dict/none | none             | Hatch pattern (style string or (style: ..., spacing: ..., stroke: ...) dict) |
| hatch-spacing | length        | 5pt              | Hatch line spacing                                                           |
| hatch-stroke  | stroke        | luma(80) + 0.5pt | Hatch line stroke                                                            |
| samples       | int           | 80               | Sample count                                                                 |

## 7.10. fill-closed - Fill a Parametric Closed Curve

Fill the interior of a closed parametric curve (the curve should start and end at the same point):

```
#plot(
  xmin: -3, xmax: 3, ymin: -2, ymax: 2,
  fill-closed(t => 2 * calc.cos(t), t => calc.sin(t),
    domain: (0, 2 * calc.pi), color: blue.lighten(70%)),
)
```

```
parametric(t => 2 * calc.cos(t), t => calc.sin(t),
           domain: (0, 2 * calc.pi), stroke: blue + 1.5pt),
)
```

#### Parameters:

| Parameter     | Type          | Default          | Description                                                               |
|---------------|---------------|------------------|---------------------------------------------------------------------------|
| fn-x          | function      | —                | X coordinate as function of $t$                                           |
| fn-y          | function      | —                | Y coordinate as function of $t$                                           |
| domain        | array         | (0.0, 1.0)       | Parameter range; curve should be closed                                   |
| color         | color         | luma(220)        | Fill color; none for hatch-only                                           |
| hatch         | str/dict/none | none             | Hatch pattern (style string or (style: .., spacing: .., stroke: ..) dict) |
| hatch-spacing | length        | 5pt              | Hatch line spacing                                                        |
| hatch-stroke  | stroke        | luma(80) + 0.5pt | Hatch line stroke                                                         |
| samples       | int           | 80               | Sample count                                                              |

### 7.11. vline / hline - Reference Lines

Draw vertical or horizontal reference lines at a fixed coordinate:

```
#plot(
  xmin: -3, xmax: 3, ymin: -2, ymax: 4,
  (fn: x => x * x - 1, stroke: blue + 1.5pt),
  vline(1.0, stroke: red + 0.8pt + (dash: "dashed")),
  hline(0.0, stroke: gray + 0.8pt),
)
```

Use ymin/ymax (for vline) or xmin/xmax (for hline) to draw partial lines:

```
vline(2.0, ymin: 0, ymax: 4)           // vertical from y=0 to y=4
hline(1.5, xmin: 0, xmax: 3)          // horizontal from x=0 to x=3
```

#### vline parameters:

| Parameter | Type       | Default           | Description                         |
|-----------|------------|-------------------|-------------------------------------|
| x0        | float      | —                 | X position of the line              |
| stroke    | stroke     | luma(100) + 0.6pt | Line stroke                         |
| ymin      | float/auto | auto              | Bottom y limit (default: plot ymin) |
| ymax      | float/auto | auto              | Top y limit (default: plot ymax)    |

#### hline parameters:

| Parameter | Type       | Default           | Description                       |
|-----------|------------|-------------------|-----------------------------------|
| y0        | float      | —                 | Y position of the line            |
| stroke    | stroke     | luma(100) + 0.6pt | Line stroke                       |
| xmin      | float/auto | auto              | Left x limit (default: plot xmin) |

|      |            |      |                                    |
|------|------------|------|------------------------------------|
| xmax | float/auto | auto | Right x limit (default: plot xmax) |
|------|------------|------|------------------------------------|

## 7.12. note - Text Annotation

Place a text annotation at a data-coordinate position:

```
#plot(
  xmin: -3, xmax: 3, ymin: -2, ymax: 4,
  (fn: x => x * x - 1, stroke: blue + 1.5pt),
  note([f(x) = x^2 - 1], pos: (1.5, 2.5), anchor: "west", size: 9pt),
)
```

### Parameters:

| Parameter | Type    | Default  | Description                |
|-----------|---------|----------|----------------------------|
| body      | content | —        | Annotation text or content |
| pos       | array   | —        | (x, y) in data coordinates |
| anchor    | string  | “center” | Text anchor point          |
| size      | length  | 9pt      | Font size                  |

### 7.13. riemann-sum - Riemann Sum Rectangles

Draw left, right, midpoint, lower, or upper Riemann sum rectangles for a function. Use it as a plot item alongside the function curve or fill-area.

The five methods:

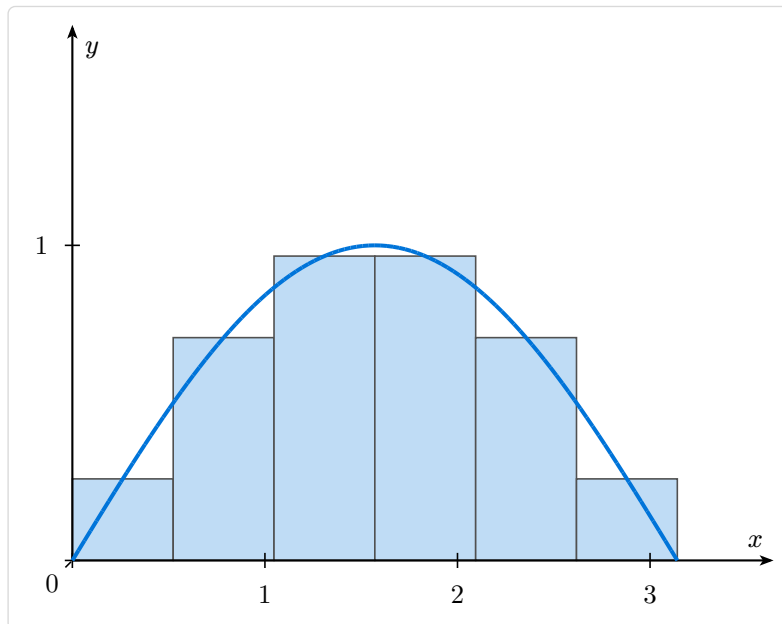
- "left" / "right" / "mid" — evaluation at the left endpoint, right endpoint, or midpoint of each sub-interval
- "lower" / "upper" — true infimum / supremum sampled within each sub-interval; correct for any function shape including U-curves

#### 7.13.1. Basic Example

Code

```
#plot(
  width: 8, height: 5,
  xmin: 0, xmax: calc.pi, ymin: 0, ymax: 1.2,
  axis-x-pos: "bottom", axis-y-pos: "left",
  riemann-sum(
    x => calc.sin(x),
    domain: (0.0, calc.pi),
    n: 6, method: "mid",
    color: blue.lighten(75%),
  ),
  (fn: x => calc.sin(x), domain: (0.0, calc.pi), stroke: blue + 1.5pt),
)
```

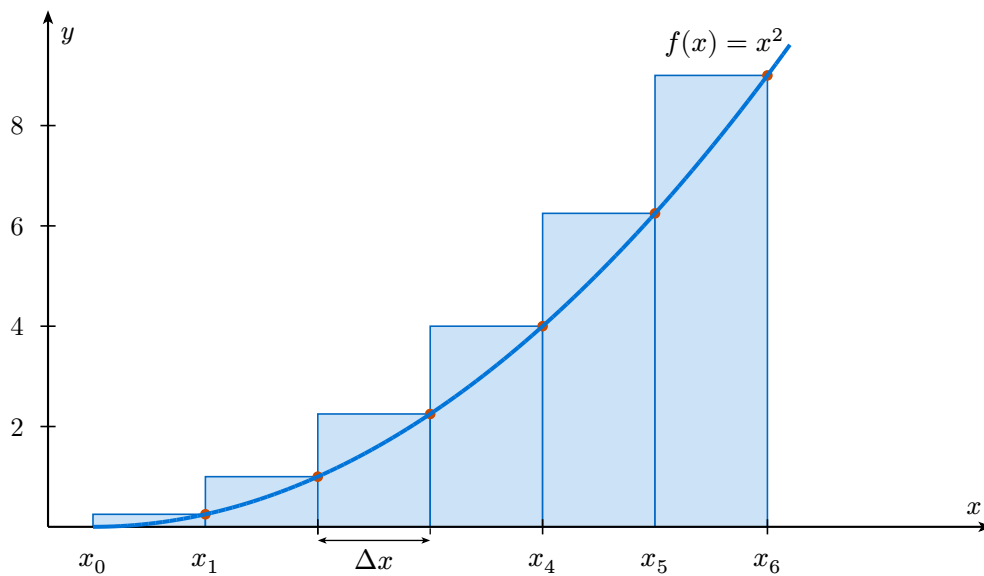
Preview



#### 7.13.2. Annotations: show-points, show-dx, show-xi

Turn on the annotation overlays for pedagogical diagrams:

```
riemann-sum(
  x => x * x,
  domain: (0.0, 3.0),
  n: 6, method: "right",
  color: blue.lighten(80%),
  show-points: true,           // dots at evaluation points with arrows
  show-dx: true, dx-rect: 2,   // Δx bracket under rectangle 2
  show-xi: true,               // x₀ ... x₆ labels along the axis
)
```



### 7.13.3. xi-show-values

Set `xi-show-values: true` to display the actual numeric values of the subdivision points instead of the symbolic  $x_i$  notation:

```
riemann-sum(fn, domain: (0.0, 4.0), n: 4, method: "left",
            show-xi: true, xi-show-values: true)
```

### 7.13.4. Hatch patterns

Use `hatch` together with `hatch-spacing` and `hatch-stroke` to draw hatched rectangles. Set `color: none` for hatch-only (no fill):

```
riemann-sum(fn, domain: ..., n: 6, method: "mid",
            color: none,
            hatch: "ne", hatch-spacing: 4pt, hatch-stroke: blue + 0.6pt)
```

Available hatch styles: "ne", "nw", "h", "v", "cross", "grid".

## 7.14. volume-of-revolution - Volume of revolution diagram

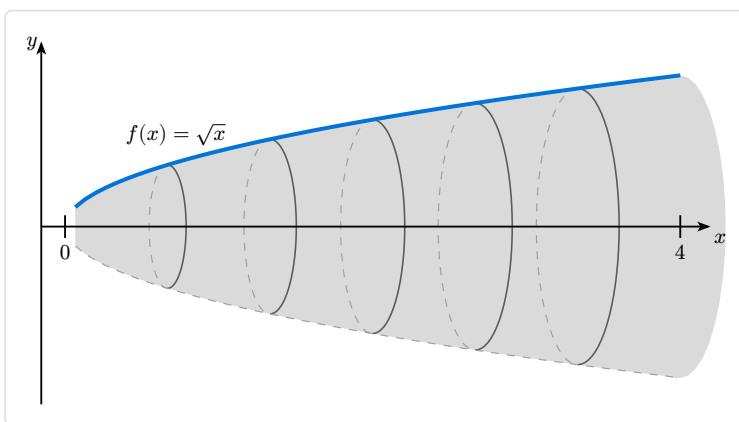
Draw a 3D-style solid generated by rotating a profile  $y = f(x)$  around a horizontal or oblique axis. The renderer draws the filled body, front and back cap ellipses, intermediate disk cross-sections, axis arrow, coordinate y-axis, and optional labels.

### 7.14.1. Basic Example

Code

```
#volume-of-revolution(  
  x => calc.sqrt(x),  
  domain: (0.0, 4.0),  
  n-disks: 5,  
  width: 8.0, height: 4.0,  
  show-y-axis: true,  
  label-a: $0$, label-b: $4$,  
  label-f: $f(x)=\sqrt{x}$$,  
)
```

Preview

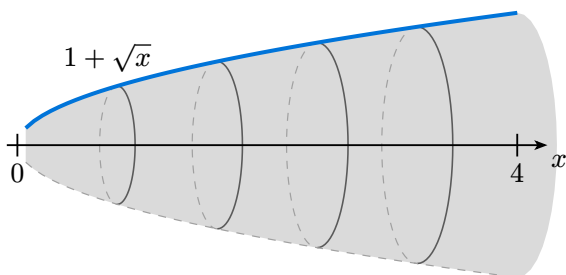


### 7.14.2. Axis of Revolution

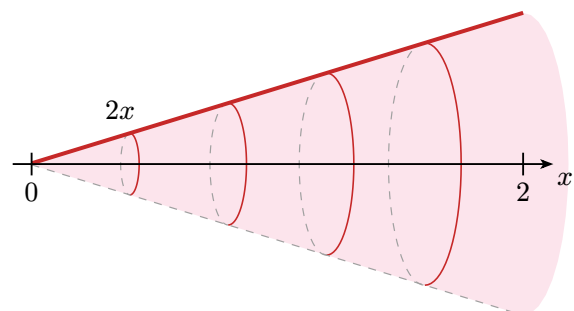
Use `axis-y` to revolve around  $y = c$  (shifted horizontal axis), or `axis-slope` for an oblique axis  $y = mx +$  `axis-y`:

```
// Around y = 1  
#volume-of-revolution(x => 1.0 + calc.sqrt(x), domain: (0, 4), axis-y: 1.0)  
  
// Around y = x (oblique)  
#volume-of-revolution(x => x * x, domain: (0.01, 2), axis-slope: 1.0)
```

*Around  $y = 1$*



*Around  $y = x$  (oblique)*



#### 7.14.3. show-back and show-radius-marker

`show-back: false` hides the dashed back half and bottom profile — useful for side-by-side comparisons. `show-radius-marker: true` draws a vertical dimension marker on the profile (at `radius-marker-x`, default: the left cap):

```
#volume-of-revolution(fn, domain: ...,  
  show-back: false,  
  show-radius-marker: true,  
  radius-marker-x: 2.0, // marker at x=2 instead of the left cap  
  label-y:  $\sqrt{2}$ ,  
)
```

#### 7.14.4. show-y-axis and y-axis placement

The coordinate y-axis is positioned automatically to the left of the solid by default. Use `y-axis-x` to fix it at a specific x-value, `y-axis-offset` to change the automatic gap, and `y-axis-extend` to control how far the axis extends above and below the solid:

```
#volume-of-revolution(fn, domain: ...,  
  show-y-axis: true,  
  y-axis-offset: 0.6, // more gap from the volume  
  y-axis-extend: (0.2, 0.5), // tighter below, more above  
)
```

## 8. Global Configuration

### 8.1. Setting Defaults

Use `set-plot-defaults` to configure defaults for all subsequent plots. Any `plot` parameter can be used, including `style`; unknown option names raise an error (typo protection):

```
#set-plot-defaults(  
  width: 8,  
  height: 6,  
  show-grid: "both",  
  minor-grid-step: 5,  
  style: (  
    axis: (stroke: navy + 0.8pt, arrow: (symbol: "stealth", fill: navy, scale: 0.55)),  
    ticks: (stroke: navy + 0.6pt, label-fill: navy),  
  ),  
)  
  
// All subsequent plots will use these defaults  
#plot(xmin: -3, xmax: 3, ymin: -2, ymax: 2, ...)
```

Per-call `style`: arguments are merged on top of the default style, so you can override individual properties without restating the full style.

### 8.2. Resetting Defaults

Reset to original defaults:

```
#reset-plot-defaults()
```

## 9. Styling

### 9.1. Custom Styles

Override default styles with the `style` parameter:

```
#plot(  
  xmin: -3, xmax: 3, ymin: -2, ymax: 2,  
  style: (  
    background: (fill: rgb("#202124")),  
    axis: (  
      stroke: white + 1pt,  
      arrow: (symbol: "stealth", fill: white, scale: 0.55),  
    ),  
    grid: (  
      major: (stroke: luma(120) + 0.6pt),  
      minor: (stroke: luma(80) + 0.3pt),  
    ),  
    ticks: (  
      length: 0.12,  
      stroke: white + 0.6pt,  
      label-size: 0.7em,  
      label-fill: white,  
    ),  
    labels: (  
      fill: white,  
    ),  
  ),  
  ...  
)
```

### 9.2. Default Style Values

| Property           | Default                                             | Description                 |
|--------------------|-----------------------------------------------------|-----------------------------|
| background.fill    | none                                                | Plot background fill        |
| background.stroke  | none                                                | Plot background stroke      |
| axis.stroke        | black + 0.8pt                                       | Axis line style             |
| axis.arrow         | (symbol: "stealth",<br>fill: black, scale:<br>0.55) | Arrow head style            |
| grid.major.stroke  | luma(200) + 0.5pt                                   | Major grid line style       |
| grid.minor.stroke  | luma(230) + 0.3pt                                   | Minor grid line style       |
| ticks.length       | 0.1                                                 | Tick mark length (cm)       |
| ticks.stroke       | black + 0.6pt                                       | Tick mark style             |
| ticks.label-size   | 0.65em                                              | Tick label font size        |
| ticks.label-fill   | black                                               | Tick label color            |
| ticks.label-offset | 0.15                                                | Distance from tick to label |
| plot.stroke        | blue + 1.2pt                                        | Default function stroke     |
| plot.samples       | 100                                                 | Default sample count        |
| marker.size        | 0.12                                                | Default marker size         |
| labels.size        | 0.8em                                               | Axis label font size        |

---

|                          |                    |                  |
|--------------------------|--------------------|------------------|
| <code>labels.fill</code> | <code>black</code> | Axis label color |
|--------------------------|--------------------|------------------|

---

## 10. Parameter Reference

### 10.1. plot Function

All parameters are named and optional; series specs are passed positionally. Unknown named arguments raise an error (typo protection).

#### Dimensions and Bounds:

| Parameter | Type         | Default | Description                                              |
|-----------|--------------|---------|----------------------------------------------------------|
| width     | length/float | 6       | Plot width — a length (2cm, 30mm, ...) or a number in cm |
| height    | length/float | 6       | Plot height — a length or a number in cm                 |
| scale     | float        | 1       | Scale factor for entire plot                             |
| xmin      | float        | −5      | Minimum x value                                          |
| xmax      | float        | 5       | Maximum x value                                          |
| ymin      | float        | −5      | Minimum y value                                          |
| ymax      | float        | 5       | Maximum y value                                          |

#### Axis Configuration:

| Parameter     | Type                  | Default      | Description                                             |
|---------------|-----------------------|--------------|---------------------------------------------------------|
| xlabel        | content               | $x$          | X-axis label                                            |
| ylabel        | content               | $y$          | Y-axis label                                            |
| xlabel-pos    | string/array          | “end”        | “end”, “center”, or (x, y)                              |
| ylabel-pos    | string/array          | “end”        | “end”, “center”, or (x, y)                              |
| xlabel-anchor | string                | “north”      | Anchor for x label                                      |
| ylabel-anchor | string                | “east”       | Anchor for y label                                      |
| xlabel-offset | array                 | (0.0, −0.05) | X label offset (cm)                                     |
| ylabel-offset | array                 | (−0.05, 0.0) | Y label offset (cm)                                     |
| axis-x-pos    | string/float/<br>none | 0            | “bottom”, “center”, a y-value, or none to hide the axis |
| axis-y-pos    | string/float/<br>none | 0            | “left”, “center”, an x-value, or none to hide the axis  |
| axis-x-extend | float/array           | (0, 0.5)     | X-axis extension (left, right)                          |
| axis-y-extend | float/array           | (0, 0.5)     | Y-axis extension (bottom, top)                          |

#### Tick Configuration:

| Parameter  | Type       | Default | Description             |
|------------|------------|---------|-------------------------|
| xtick      | array/none | auto    | Custom x tick positions |
| ytick      | array/none | auto    | Custom y tick positions |
| xtick-step | float      | 1       | X tick spacing          |
| ytick-step | float      | 1       | Y tick spacing          |

|                                       |            |                |                                                                                                                                                        |
|---------------------------------------|------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>xtick-labels</code>             | array/none | auto           | Custom x tick labels                                                                                                                                   |
| <code>ytick-labels</code>             | array/none | auto           | Custom y tick labels                                                                                                                                   |
| <code>xtick-label-step</code>         | int        | 1              | Show x label every N ticks                                                                                                                             |
| <code>ytick-label-step</code>         | int        | 1              | Show y label every N ticks                                                                                                                             |
| <code>show-origin</code>              | bool       | true           | Show “0” at origin                                                                                                                                     |
| <code>origin-label-offset</code>      | array      | (-0.11, -0.11) | Origin “0” label offset from (0,0) in cm                                                                                                               |
| <code>origin-label-anchor</code>      | string     | “north-east”   | Origin “0” label anchor                                                                                                                                |
| <code>origin-leader</code>            | bool       | true           | Draw a subtle leader line from origin label toward (0,0)                                                                                               |
| <code>origin-leader-stroke</code>     | stroke     | black + 0.6pt  | Origin leader stroke                                                                                                                                   |
| <code>origin-leader-gap</code>        | float      | 0.025          | Gap from (0,0) before leader starts (cm)                                                                                                               |
| <code>origin-leader-end-gap</code>    | float      | 0.025          | Gap before the label anchor (cm)                                                                                                                       |
| <code>unit-label-only</code>          | bool       | false          | Show only “1” on axes                                                                                                                                  |
| <code>tick-label-size</code>          | length     | 0.65em         | Tick label font size                                                                                                                                   |
| <code>axis-label-size</code>          | length     | 0.8em          | Axis label font size                                                                                                                                   |
| <code>min-tick-spacing</code>         | float      | 0.4            | Minimum distance between ticks (cm); auto-widens step to the next “nice” value (1, 2, 5, 10...) when the range is large                                |
| <code>hide-crossed-tick-labels</code> | bool       | true           | Hide individual tick labels that a plotted curve crosses, keeping labels legible                                                                       |
| <code>show-end-ticks</code>           | bool       | true           | Keep the tick/label at <code>xmax/ymax</code> when it lands on a tick (e.g. show “5”), pushing the axis slightly past it so the arrow clears the label |

### Grid Configuration:

| Parameter        | Type        | Default | Description                           |
|------------------|-------------|---------|---------------------------------------|
| show-grid        | bool/string | false   | true, false, “major”, “minor”, “both” |
| minor-grid-step  | int         | 5       | Subdivisions per major tick           |
| grid-label-break | bool        | true    | Break grid lines around labels        |

### Styling:

| Parameter | Type       | Default | Description                                                                                                          |
|-----------|------------|---------|----------------------------------------------------------------------------------------------------------------------|
| style     | dictionary | none    | Style overrides (see Styling section)                                                                                |
| font      | str/array  | auto    | Font applied to all plot-generated text (tick labels, axis labels, origin, annotations); <b>auto</b> = document font |
| series    | array      | none    | Pre-built array of series specs (alternative to positional args)                                                     |

## 10.2. Function/Data Specification

Each plot item is a dictionary with these fields:

| Field        | Type     | Description                                                                                          |
|--------------|----------|------------------------------------------------------------------------------------------------------|
| fn           | function | Function to plot: $x \Rightarrow y$ (return <b>none</b> to create a hole/discontinuity)              |
| data         | array    | Data points: $((x_1, y_1), (x_2, y_2), \dots)$                                                       |
| connect      | bool     | Connect data points with a line (default: <b>true</b> for <b>fn</b> , <b>false</b> for <b>data</b> ) |
| domain       | array    | Function domain: $(x_{\min}, x_{\max})$                                                              |
| samples      | int      | Number of samples for function (default: 100)                                                        |
| stroke       | stroke   | Line style (default: <b>blue + 1.2pt</b> )                                                           |
| mark         | string   | Marker type (default: “none”)                                                                        |
| mark-size    | float    | Marker size in cm (default: 0.12)                                                                    |
| mark-fill    | color    | Marker fill color (default: black)                                                                   |
| mark-stroke  | stroke   | Marker stroke style (default: <b>black + 0.8pt</b> )                                                 |
| label        | content  | Label text placed near the curve                                                                     |
| label-pos    | float    | Position along the curve in $[0, 1]$ (default: 1.0)                                                  |
| label-side   | string   | Label placement relative to curve: “above”, “below”, “left”, “right”, “above-left”, etc.             |
| label-anchor | string   | CeTZ anchor override (alternative to <b>label-side</b> )                                             |

### 10.3. riemann-sum Function

| Parameter       | Type              | Default          | Description                                                                                              |
|-----------------|-------------------|------------------|----------------------------------------------------------------------------------------------------------|
| fn              | function          | —                | Function $f(x)$ to approximate (positional or fn:)                                                       |
| domain          | array             | plot range       | Interval (a, b); overrides the plot's x-range                                                            |
| n               | int               | 4                | Number of rectangles                                                                                     |
| method          | string            | "right"          | "left", "right", "mid", "lower", "upper"                                                                 |
| baseline        | float             | 0.0              | Y-level of rectangle bases (default: x-axis)                                                             |
| color           | color             | luma(220)        | Rectangle fill; none for hatch-only                                                                      |
| stroke          | stroke            | luma(80) + 0.6pt | Rectangle border stroke                                                                                  |
| hatch           | str/dict/none     | none             | Hatch pattern: "ne", "nw", "h", "v", "cross", "grid" — or (style: "ne", spacing: 5pt, stroke: red + 1pt) |
| hatch-spacing   | length            | 5pt              | Spacing between hatch lines                                                                              |
| hatch-stroke    | stroke            | luma(80) + 0.5pt | Stroke for hatch lines                                                                                   |
| samples         | int               | 20               | Samples per sub-interval for "lower" / "upper" methods                                                   |
| show-points     | bool              | false            | Draw a dot at each evaluation point                                                                      |
| point-color     | color             | rgb("#c94a00")   | Dot fill color                                                                                           |
| point-size      | float             | 0.07             | Dot radius in cm                                                                                         |
| point-label     | content/auto/none | auto             | Arrow label near dots; auto = method name, none = dots only                                              |
| point-label-pos | array/auto        | auto             | (x, y) in data coords; auto = upper-right of dots                                                        |
| show-dx         | bool              | false            | Draw a $\Delta x$ dimension bracket under one rectangle                                                  |
| dx-rect         | int/auto          | auto             | Rectangle to annotate (0-based); auto = middle rectangle                                                 |
| dx-label        | content           | $\Delta x$       | Label displayed inside the bracket                                                                       |
| show-xi         | bool              | false            | Draw $x_0, x_1, \dots, x_n$ at subdivision points                                                        |
| xi-labels       | array/auto        | auto             | Custom label array; auto = subscripted $x_i$ notation                                                    |
| xi-show-values  | bool              | false            | Show numeric x-values instead of $x_i$ notation                                                          |

## 10.4. volume-of-revolution Function

`solid-of-revolution` is an alias kept for backward compatibility.

| Parameter                       | Type         | Default                       | Description                                                                                                 |
|---------------------------------|--------------|-------------------------------|-------------------------------------------------------------------------------------------------------------|
| <code>fn</code>                 | function     | —                             | Profile function $f(x)$ ; the curve that is revolved                                                        |
| <code>domain</code>             | array        | <code>(0.0, 4.0)</code>       | Interval <code>(a, b)</code> — the revolution interval                                                      |
| <code>n-disks</code>            | int          | 4                             | Number of intermediate disk cross-sections to show                                                          |
| <code>width / height</code>     | length/float | 5.0 / 3.5                     | Canvas size — a length or a number in cm                                                                    |
| <code>samples</code>            | int          | 60                            | Number of points sampled along the profile curve                                                            |
| <code>axis-y</code>             | float        | 0.0                           | Y-value of horizontal revolution axis ( $y = c$ )                                                           |
| <code>axis-slope</code>         | float        | 0.0                           | Slope $m$ : revolution axis is $y = mx + \text{axis-y}$                                                     |
| <code>show-axis</code>          | bool         | true                          | Draw the revolution axis with an arrowhead                                                                  |
| <code>show-y-axis</code>        | bool         | false                         | Draw a coordinate y-axis at the left cap ( <code>show-yaxis</code> is a deprecated alias)                   |
| <code>y-axis-x</code>           | float/auto   | <code>auto</code>             | X-position for the y-axis; <code>auto</code> = left of volume                                               |
| <code>y-axis-offset</code>      | float        | 0.45                          | Gap between y-axis and the volume when <code>auto</code>                                                    |
| <code>y-axis-extend</code>      | array        | <code>(0.35, 0.45)</code>     | Y-axis padding (below, above) the solid                                                                     |
| <code>show-radius-marker</code> | bool         | false                         | Draw a vertical radius dimension marker at <code>radius-marker-x</code>                                     |
| <code>radius-marker-x</code>    | float/auto   | <code>auto</code>             | X-position of the radius marker; <code>auto</code> = left cap ( <code>yaxis-x</code> is a deprecated alias) |
| <code>show-back</code>          | bool         | true                          | Show the back half of the solid (dashed ellipses + bottom curve)                                            |
| <code>show-labels</code>        | bool         | true                          | Show $a$ , $b$ , $f$ labels                                                                                 |
| <code>profile-stroke</code>     | stroke       | <code>blue + 1.5pt</code>     | Top profile curve stroke                                                                                    |
| <code>disk-color</code>         | color        | <code>luma(218)</code>        | Solid body fill color                                                                                       |
| <code>disk-stroke</code>        | stroke       | <code>luma(90) + 0.6pt</code> | Disk edge stroke                                                                                            |
| <code>axis-stroke</code>        | stroke       | <code>black + 0.7pt</code>    | Revolution axis stroke                                                                                      |
| <code>label-a / label-b</code>  | content      | <code>\$a\$ / \$b\$</code>    | Labels at domain endpoints                                                                                  |

|         |         |     |                                       |
|---------|---------|-----|---------------------------------------|
| label-f | content | $f$ | Function label near the profile curve |
| label-y | content | $y$ | Label for the coordinate y-axis       |

## 10.5. zoom - Spy / zoom inset

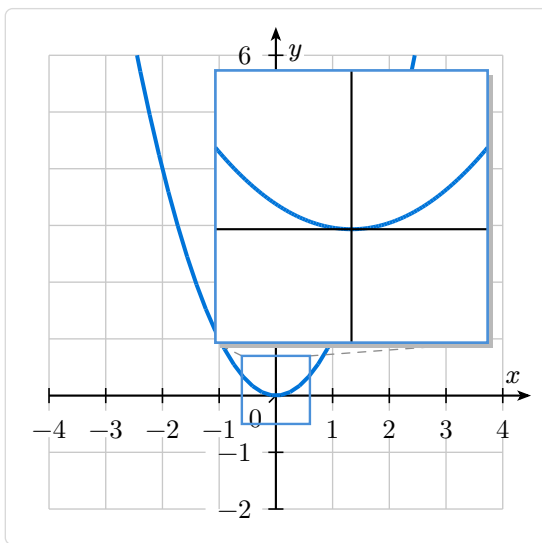
Add a magnified sub-view (spy glass) of any region of the main plot. The `zoom()` helper returns a series spec that is passed directly to `plot()`. It draws a highlighted region on the main plot and a zoomed inset box with connector lines. All series types are re-rendered inside the inset — curves, points, fills, Riemann rectangles, reference lines and parametric curves — clipped to the lens shape (annotations and labels are not repeated).

### 10.5.1. Basic Example

Code

```
#plot(  
  xmin: -4, xmax: 4, ymin: -2, ymax: 6,  
  show-grid: "major",  
  (fn: x => x * x, stroke: blue + 1.5pt),  
  zoom(center: (0, 0.1), size: 0.9, magnification: 4, at: "top-right"),  
)
```

Preview

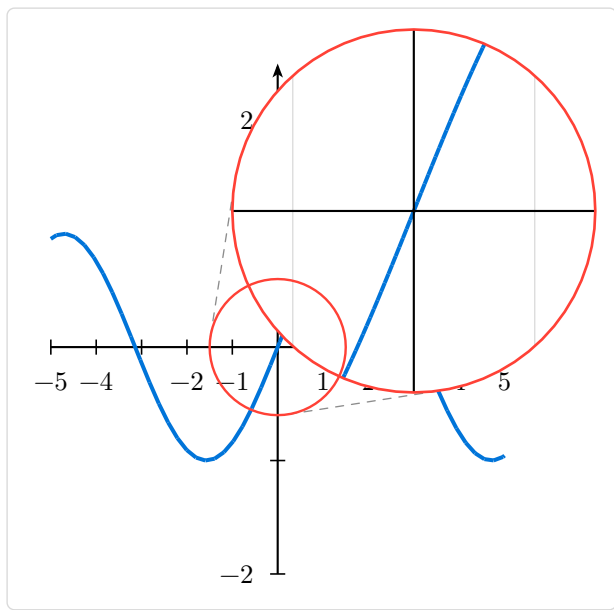


### 10.5.2. Circular lens, explicit region

Code

```
#plot(
  xmin: -5, xmax: 5, ymin: -2, ymax: 2,
  (fn: x => calc.sin(x), stroke: blue + 1.5pt),
  zoom(region: (-0.5, -0.6, 0.5, 0.6),
    lens-shape: "circle", magnification: 5,
    at: (3, 1.2), accent: red),
)
```

Preview



### 10.5.3. Parameters

| Parameter      | Type         | Default | Description                                                                                                                                                                                     |
|----------------|--------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| region         | array        | auto    | Spy glass corners (x1, y1, x2, y2) in data coords. Either region or center + size is required.                                                                                                  |
| center         | array        | auto    | Spy glass center (cx, cy) in data coords (use with size)                                                                                                                                        |
| size           | length/float | auto    | Spy glass size (length, or number in cm) — ensures a square glass regardless of axis ratio                                                                                                      |
| at             | array/str    | auto    | Inset box center: (x,y) data coords, or a placement keyword: "top-right", "top-left", "bottom-right", "bottom-left", "top", "bottom", "left", "right". auto = smart opposite-quadrant placement |
| width / height | length/float | auto    | Inset box size (length, or number in cm). auto = derived from magnification or 3.5                                                                                                              |
| magnification  | float        | auto    | Zoom factor — inset size = spy glass canvas size × factor                                                                                                                                       |
| lens-shape     | str          | "rect"  | "rect" or "circle"                                                                                                                                                                              |
| connect        | bool         | true    | Draw connector lines between spy glass and inset                                                                                                                                                |
| accent         | color        | blue    | Accent color for borders and region highlight                                                                                                                                                   |
| region-fill    | color        | auto    | Fill tint inside the spy glass region                                                                                                                                                           |
| region-stroke  | stroke       | auto    | Border of the spy glass region                                                                                                                                                                  |
| box-fill       | color        | white   | Background fill of the inset box                                                                                                                                                                |
| box-stroke     | stroke       | auto    | Border of the inset box                                                                                                                                                                         |

|                    |         |       |                                                                |
|--------------------|---------|-------|----------------------------------------------------------------|
| connector-stroke   | stroke  | auto  | Dashed gray connector lines by default                         |
| connector-fill     | color   | none  | Fill the trapezoid between connector lines                     |
| shadow             | bool    | auto  | Drop shadow behind the inset (default true for rect with fill) |
| show-inset-grid    | bool    | true  | Subtle grid inside the inset aligned to main ticks             |
| show-magnification | bool    | false | Show $\times N$ label in inset corner                          |
| label              | content | none  | Custom label inside the inset (overrides the $\times N$ label) |

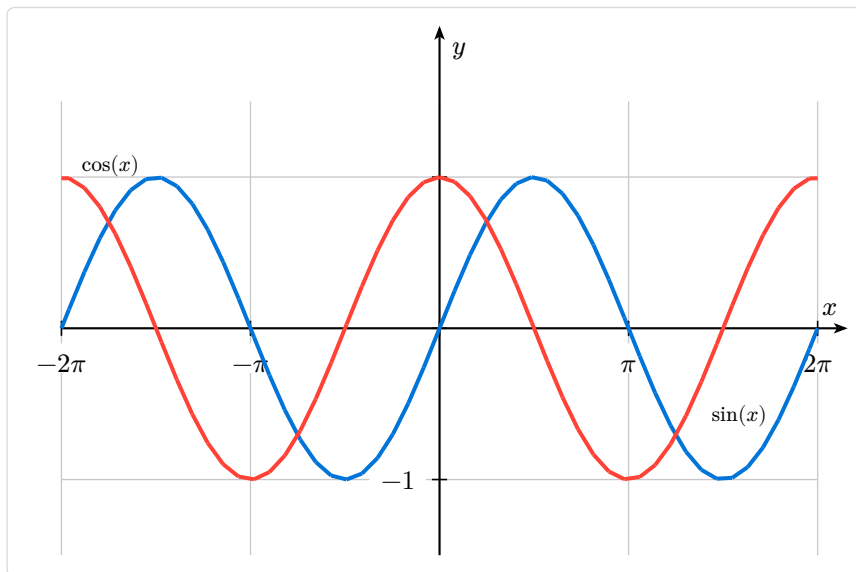
## 11. Complete Examples

### 11.1. Trigonometric Functions

Code

```
#plot(  
  width: 10, height: 6,  
  xmin: -2 * calc.pi, xmax: 2 * calc.pi,  
  ymin: -1.5, ymax: 1.5,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  xtick: (-2*calc.pi, -calc.pi, 0, calc.pi, 2*calc.pi),  
  xtick-labels: ( $-2\pi$ ,  $-\pi$ ,  $0$ ,  $\pi$ ,  $2\pi$ ),  
  show-grid: "major",  
  (fn: x => calc.sin(x), stroke: blue + 1.5pt, label:  $\sin(x)$ , label-pos: 1),  
  (fn: x => calc.cos(x), stroke: red + 1.5pt, label:  $\cos(x)$ , label-pos: 0),  
)
```

Preview

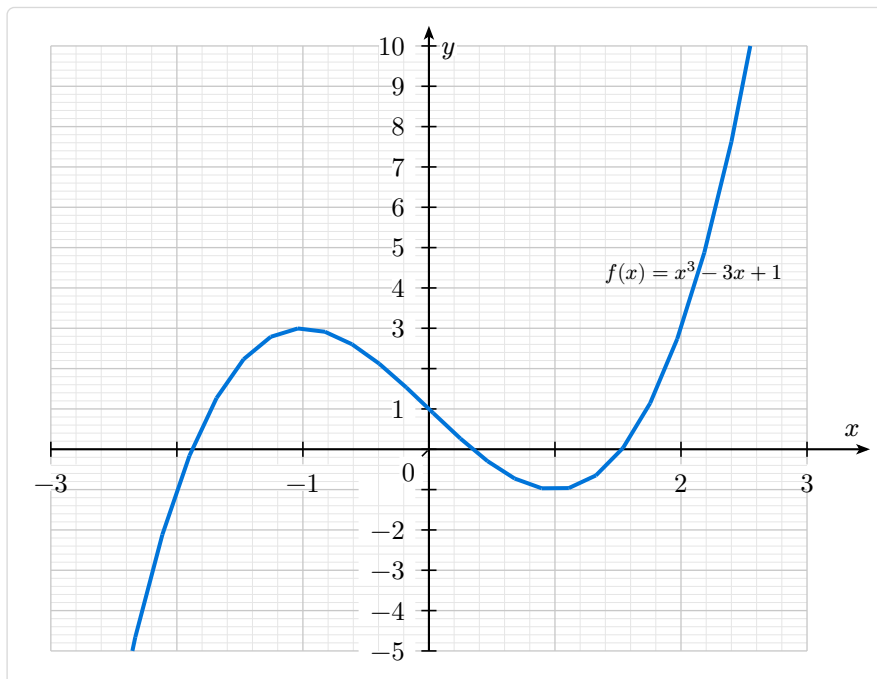


## 11.2. Polynomial with Fine Grid

Code

```
#plot(  
  width: 10, height: 8,  
  xmin: -3, xmax: 3, ymin: -5, ymax: 10,  
  xlabel: $x$, ylabel: $y$,  
  show-grid: "both",  
  minor-grid-step: 5,  
  (  
    fn: x => x * x * x - 3 * x + 1,  
    stroke: blue + 1.5pt,  
    label: $f(x) = x^3 - 3x + 1$,  
    label-side: "above",  
    label-pos: 0.85,  
  ),  
)
```

Preview

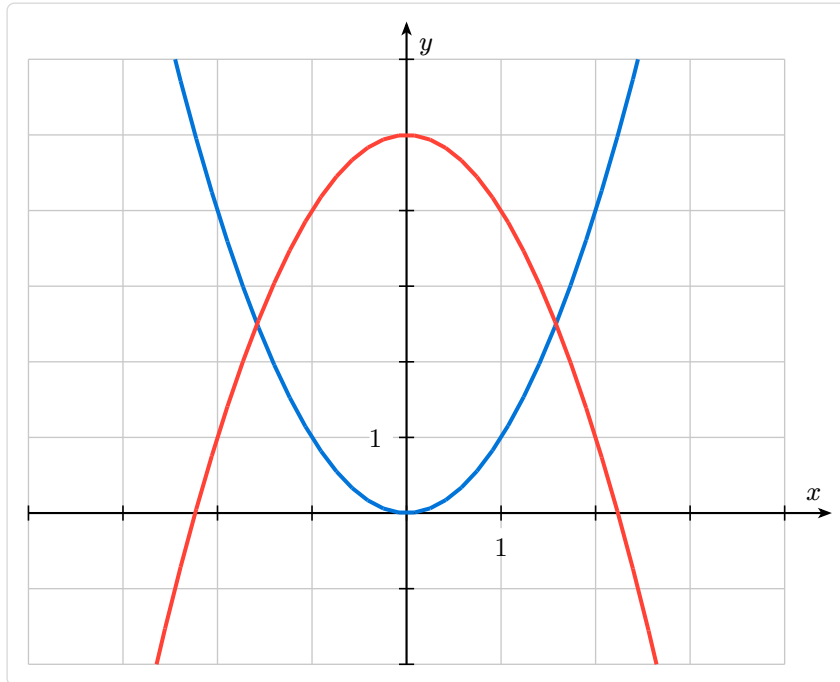


## 11.3. Minimal Style Plot

Code

```
#plot(  
  width: 10, height: 8,  
  xmin: -4, xmax: 4, ymin: -2, ymax: 6,  
  xlabel:  $x$ , ylabel:  $y$ ,  
  show-grid: "major",  
  unit-label-only: true,  
  show-origin: false,  
  (fn: x => x * x, stroke: blue + 1.5pt),  
  (fn: x => -x * x + 5, stroke: red + 1.5pt),  
)
```

Preview



## 11.4. Data with Trend Line

Code

```
#plot(
  width: 10, height: 7,
  xmin: 0, xmax: 6, ymin: 0, ymax: 12,
  xlabel: "Time (s)", ylabel: "Distance (m)",
  axis-x-pos: "bottom", axis-y-pos: "left",
  show-grid: "both",
  minor-grid-step: 5,
  (fn: x => 2 * x, stroke: gray + 1pt, domain: (0, 6)), // Trend line
  (
    data: ((0.5, 1.2), (1, 2.3), (2, 3.8), (3, 6.2), (4, 7.9), (5, 10.1)),
    mark: "o",
    mark-size: 0.12,
    stroke: none,
  ),
)
```

Preview

